

Chapter 10 — Statistical distributions and simulation

Before deep generative models, teams synthesized data with explicit statistical assumptions

Learning objectives

- By the end of this part

Statistical methods overview

- Statistical synthesis uses mathematical models to simulate data that shares properties
- Common tools include distribution fitting, Monte Carlo draws, bootstrapping
- The main challenge is matching correlations and trends, not just marginal distributions

Data distribution modeling

- Distribution modeling fits a known law
- Draw synthetic values from the same family
- This works when the assumed distribution matches the domain

Example 10.6 — Normal-Distribution Sampling Sketch

- Example 10.6 — hands-on module
- Example 10.6 sketches sampling from a fitted normal distribution
- If real values follow a mean of fifty and standard deviation of ten
- Explore the chapter example module
- View files: ``modules/chapter10/example6/``

Example 10.6 — listing

```
"""Example 10.6 – sample synthetic points from a fitted normal
distribution."""

from __future__ import annotations

import random
from statistics import fmean, pstdev

def main() -> None:
    """Draw samples from  $N(50, 10^2)$  and compare sample moments to the
    target."""
    random.seed(10)
    mean, sd, n = 50.0, 10.0, 1000
    samples = [random.gauss(mean, sd) for _ in range(n)]
    print(f"Target distribution:  $N(\text{mean}=\{\text{mean}\}, \text{sd}=\{\text{sd}\})$ ")
    print(f"Sample size: {n}")
    print(f"Sample mean: {fmean(samples):.2f}")
    print(f"Sample std dev: {pstdev(samples):.2f}")
```

Monte Carlo simulation

- Monte Carlo methods repeat random sampling and simulation when the underlying process is
- Many draws are averaged or aggregated to estimate outcomes under uncertainty
- Finance, engineering

Example 10.7 — Monte Carlo Asset-Return Simulation

- Example 10.7 — hands-on module
- Example 10.7 applies Monte Carlo simulation to financial returns
- Synthetic return paths incorporate risk factors and market conditions
- Explore the chapter example module
- View files: ``modules/chapter10/example7/``

Example 10.7 — listing

```
"""Example 10.7 – Monte Carlo simulation of asset returns under
uncertainty."""

from __future__ import annotations

import random
from statistics import fmean, pstdev

def main() -> None:
    """Simulate many one-period returns and summarize the resulting
    distribution."""
    random.seed(10)
    # Annualized mean return and volatility (illustrative).
    mu, sigma = 0.08, 0.20
    n_paths = 5_000
    returns = [random.gauss(mu, sigma) for _ in range(n_paths)]
```

Controlled simulation benchmarks

- Some releases, such as device-model sweeps, are synthetic by construction
- Quality assurance then documents settings and reproducibility rather than detecting
- That differs from privacy-motivated tabular synthesis but still produces shareable

Takeaways

- Parametric sampling copies moments from fitted laws
- Both require explicit assumptions and validation against real benchmarks
- Univariate methods ignore variable relationships

Next

- Complete the quiz for this part
- Regression-based projection for structured synthetic rows